University of Eloued

Faculty of economy nag management science

Open Statisical softwars

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Title: Exercises on Lag Selection and Causality Tests Using R

Introduction

In time series econometrics, the study of dynamic relationships between macroeconomic variables often requires the use of lag structures and causality tests. Vector Autoregressive (VAR) models provide a flexible framework to analyze interactions among multiple time series variables. Using statistical software such as R, researchers can estimate these models, select optimal lag lengths, and perform causality tests.

The following exercises are based on a dataset containing the variables:

Date
GDP_Index
Inflation_Rate
Interest_Rate
Exchange_Rate
Stock_Market_Index

These exercises aim to develop practical skills in time series analysis using RStudio.

Exercise 1
Title: Exploring the Dataset

Questions

1. Load the dataset into RStudio.
2. Display the names of all variables in the dataset.
3. Display the first observations of the dataset.
4. Describe the economic meaning of the following variables:
   GDP_Index, Inflation_Rate, Interest_Rate, Exchange_Rate, and Stock_Market_Index.

Tasks in R

Use appropriate R commands to:

• Display variable names
• Display the first rows of the dataset
• Summarize the dataset

Exercise 2
Title: Visualization of Macroeconomic Time Series

Questions

1. Plot the time series of GDP_Index.
2. Plot the time series of Inflation_Rate.
3. Plot the time series of Exchange_Rate.
4. Based on the graphs, discuss whether the variables appear to exhibit trends or structural changes.

Tasks in R

Use R graphical functions to generate the required time series plots.

Exercise 3
Title: Testing for Stationarity

Questions

1. Explain the importance of stationarity in time series modeling.
2. Perform the Augmented Dickey–Fuller (ADF) test for the following variables:

GDP_Index
Exchange_Rate
Interest_Rate

3. State the null hypothesis of the ADF test.
4. Based on the test results, determine whether each series is stationary.

Tasks in R

Use appropriate R packages and functions to perform the stationarity tests.

Exercise 4
Title: Transformation of Non-Stationary Series

Questions

1. Compute the first difference of the following variables:

Exchange_Rate
Interest_Rate

2. Create two new variables named:

Exchange_Rate_diff
Interest_Rate_diff

3. Construct a new dataset containing the transformed variables.
4. Plot the differenced series and compare them with the original series.

Tasks in R

Use appropriate commands to compute the first differences and create a new dataframe.

Exercise 5
Title: Lag Length Selection in VAR Models

Questions

1. Explain the role of lag length in Vector Autoregressive models.
2. Using the differenced variables Exchange_Rate_diff and Interest_Rate_diff, determine the optimal lag length.
3. Compare the results obtained from the following criteria:

AIC (Akaike Information Criterion)
HQ (Hannan–Quinn Criterion)
SC (Schwarz Criterion)
FPE (Final Prediction Error)

4. Discuss how a researcher should decide which lag length to use when different criteria suggest different values.

Tasks in R

Use the appropriate function to perform lag selection for the VAR model.

Exercise 6
Title: Estimation of a VAR Model

Questions

1. Estimate a Vector Autoregressive model using the variables:

Exchange_Rate_diff
Interest_Rate_diff

2. Use the optimal lag length selected in the previous exercise.
3. Present the estimated coefficients of the model.
4. Explain the economic meaning of lagged coefficients in a VAR framework.

Tasks in R

Estimate the VAR model and display the results.

Exercise 7
Title: Granger Causality Test

Questions

1. Explain the concept of Granger causality in time series econometrics.
2. State the null hypothesis for testing whether Exchange_Rate_diff Granger-causes Interest_Rate_diff.
3. Perform the Granger causality test using the estimated VAR model.
4. Interpret the statistical significance of the test results.

Tasks in R

Use the appropriate function to conduct the causality test within the VAR framework.

Exercise 8
Title: Instantaneous Causality Test

Questions

1. Define instantaneous causality in the context of VAR models.
2. Explain the difference between lagged causality and instantaneous causality.
3. Perform a test for instantaneous causality between Exchange_Rate_diff and Interest_Rate_diff.
4. Discuss the economic interpretation of the results.

Tasks in R

Use the relevant function to extract instantaneous causality results.

Exercise 9
Title: Macroeconomic Interpretation of Dynamic Relationships

Questions

1. Discuss the possible relationship between inflation and interest rates.
2. Explain how exchange rate fluctuations may influence stock market performance.
3. Discuss how GDP growth might affect stock market indices.
4. Explain why macroeconomic variables often exhibit dynamic interdependence.